Roll No.....

Mid Semester Examination March 2025

Name of the Program: ***B.Tech AE*** *Semester:* **IV**

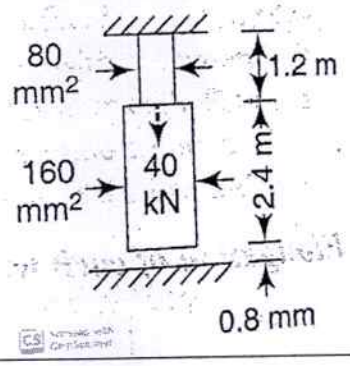
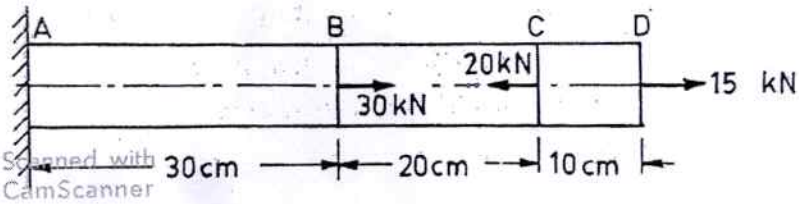
Name of the Course: **Solid Mechanics** *Course Code:* **TAS 406**

Time: 90 Minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Deduce expression to determine the elongation of a bar of trapezoidal section with uniform thickness.	
	OR	
(b)	A suspended bar system consists of two cross-sections as shown in figure. Initially its lower surface is 0.8 mm above the ground surface. Determine the reaction of the lower support and the stresses in each section when a load of 40 kN is applied as shown in the figure. Take E = 205GPa 	**CO1**
Q2	(10 marks)	
(a)	Explain the following properties: (i) Elasticity (ii) Plasticity (iii) Ductility (iv) Brittleness (v) Toughness	
	OR	
(b)	A steel bar of 30 mm diameter is loaded as shown in Figure. Determine the stress in each portion and the total elongation. E = 210 GPa. 	**CO1**

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Q3	(10 marks)	**CO1, CO2**
(a)	What do you mean by thermal stresses? Explain	
	OR	
(b)	(i) Deduce the expression for elongation of a bar of rectangular section due to self-weight. (ii) A steel wire of 8-mm diameter is used to lift a weight of 1.5 kN at its lowest end. The density of the wire material is 8000 kg/m³. Determine the elongation of the wire if the length of the wire is 100m. E=205 GPa	
Q4	(10 marks)	**CO2**
(a)	What is Mohr's stress circle? How the Mohr's circle is constructed?	
	OR	
(b)	Rails are laid such that there is no stress in them at 34° C. If the rails are 32 m long. Determine: (i) Stress in the rail at 90° C, when there is no allowance for expansion (ii) Stress in the rail at 90° C, when there is expansion allowance of 8mm per rail. (iii) The expansion allowance for no stress in the rail at 90°C. Take: Coefficient of linear expansion = 11 x 10^(-6) /°C, E= 205 GPa	
Q5	(10 marks)	**CO2**
(a)	What do you mean by principal planes and principal stresses? Derive the expression for principal stresses for a body subjected to direct and shear stresses.	
	OR	
(b)	Draw Mohr's stress circle indicating the radius and centre point coordinate for the given stress conditions. (i) Pure shear (50 MPa) (ii) Hydrostatic stresses (100 MPa) (iii) Normal stress in X- direction 100 MPa(tensile) and Normal stress in Y-direction 50 MPa(tensile) (iv) Normal stress in X- direction 100 MPa(tensile) and Normal stress in Y-direction 50 MPa(compressive) (v) Normal stress in X- direction 100 MPa(tensile) and Normal stress in Y-direction 50 MPa(tensile) with 50 MPa shear stress	